PREOPERATIVE DIAGNOSIS: Macula-sparing rhegmatogenous retinal detachment with superior horseshoe retinal tear, left eye.

POSTOPERATIVE DIAGNOSIS: Rhegmatogenous retinal detachment, left eye.

PROCEDURE PERFORMED: Left eye 25-gauge pars plana vitrectomy, posterior vitreous detachment induction, fluid-air exchange, endolaser photocoagulation, and 20% sulfur hexafluoride (SF6) gas endotamponade.

OPERATIVE STEPS: Under monitored anesthesia care supplemented by a left retrobulbar block (4 mL of 50/50 mixture of 2% lidocaine and 0.75% bupivacaine), the left periocular area was prepped and draped sterilely. A 25-gauge Alcon Constellation vitrectomy system was utilized. Three valved cannula trocars were placed transconjunctivally through the pars plana 3.5 mm posterior to the limbus in the superotemporal, superonasal, and inferotemporal quadrants.

The infusion cannula was connected and confirmed in the vitreous cavity. Core vitrectomy was executed. Triamcinolone acetonide suspension was injected to visualize the transparent vitreous cortex. Complete posterior vitreous detachment (PVD) was induced using active aspiration over the optic nerve head. Peripheral vitrectomy with 360-degree scleral depression was carried out to shave the vitreous base.

A large horseshoe retinal tear with subretinal fluid extending into the superotemporal periphery (sparing the fovea) was clearly visualized. Perfluorocarbon liquid (Perfluoron) was injected over the posterior pole to stabilize the retina and express subretinal fluid anteriorly through the break. Fluid-air exchange was performed, thoroughly aspirating all subretinal fluid until the retina was completely flattened against the retinal pigment epithelium.

Endolaser photocoagulation was applied in three confluent rows surrounding the retinal break and 360 degrees around the peripheral retina. Perfluoron was completely aspirated. An exchange of air for a non-expansile 20% SF6 gas mixture was executed. The trocars were removed; the sclerotomies were self-sealing and verified non-leaking. Subconjunctival vancomycin and dexamethasone were administered. The patient was instructed to maintain strict face-down positioning.